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Algorithmic agencing in platform markets

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Abstract

Algorithms are central in shaping platform markets, impacting not only calculation processes but also influencing market actors' relationships and market structures more broadly. As "autonomous" market devices that are executable or operate without the need for human intervention, algorithms continuously organize, prioritize, and rank product profiles, resulting in personalized exchange offers based on data processing. However, different market actors also actively attempt to influence these processes, turning platform markets into what could be described as algorithmic battlefields. Despite the growing significance of market-based algorithms, research on their role in market shaping remains scarce. This paper explores the role of algorithms in shaping platform markets. By drawing together literatures from market studies and critical algorithm studies, four key algorithmic characteristics in "agencing" markets are identified: as market actors, targets of market contestations, boundary objects, and market systems connectors. Based on these roles, the paper formulates a research agenda for studying algorithmic agencing in platform markets.

Keywords

Actor-network theory, agencing, algorithms, market shaping, market studies, platforms

Introduction

This paper investigates the agency of algorithms in shaping platform markets. In consumer-facing platform markets such as Amazon, eBay, and AirBnB, algorithms have been playing a central role in ordering and organizing market objects and actors. Among many other tasks, they rank products and sellers based on specific features, and they recommend suitable products to buyers (Zhong, 2023). While buyers tend to trust the results of these algorithms because of their appearance of rationality and objectivity (Beer, 2017), a recent controversial practice of Amazon's Buy Box has shown how

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easily this trust can be undermined. As charged by the European Commission, Amazon is believed to use its algorithms to preferentially select products from its own retail business and marketplace sellers that use Amazon's logistics and delivery services into Buy Box, a space on its website that features a sole seller's offer and enables quick product purchases through a straightforward click on the purchase button (European Commission, 2022). This example demonstrates that the driving forces behind some market failures in platform markets are not human actors but non-human market devices—algorithms. However, despite the growing importance of market-based algorithms, there is still scarce research in conceptualizing the role of algorithms in shaping markets (though see Airoldi and Rokka, 2022; Pellandini-Simányi, 2023). We propose that the lens of market studies might reveal how algorithms affect some key market dynamics, contributing an important market studies angle to the rapidly growing body of critical algorithm studies.

In particular, we focus on algorithms' agentic roles in platform markets. The rapid expansion of digital platforms in different domains has attracted a great deal of attention from market studies researchers (e.g., Caliandro et al., 2024; Drenten et al., 2024; Hagberg and Kjellberg, 2020; Scaraboto and Fischer, 2024). What these researchers have called the "platformization of markets" refers to "the penetration of infrastructures, economic processes and governmental frameworks of digital platforms in different economic sectors and spheres of life" (Poell et al., 2019). As platforms, we refer to online marketplaces that enable interactions between multiple groups of connected actors, such as platform owners and the actors on different sides of multi-sided markets (Boudreau and Hagiu, 2009). Platforms constitute "data infrastructures that facilitate, aggregate, monetize, and govern interactions between end-users and content and service providers" (Poell et al., 2021: 5). Being in control of such data infrastructures means that platform owners have the ability to influence user interactions by using algorithms to enhance or limit exposure (Scaraboto and Fischer, 2024). Indeed, algorithms have been deployed to segment or classify customers—often with significant repercussions—and recommend differentiated contents (Pellandini-Simányi, 2023). The opacity and constant evolution of platform algorithms can create uncertainty among users regarding their current operation and the impact of future updates (Duffy, 2020). For instance, platform sellers' own value-creation practices can be affected by the ever-changing logic of algorithms, as many algorithms are inscrutable "by design" (Wood and Lehdonvirta, 2021).

Algorithms are often considered autonomous because they are executable or operate without the need for human intervention (Introna, 2016); they function independently of both programming languages and the hardware on which these programs run (Goffey, 2008), and they automate decision-making processes with potentially opaque outcomes (Zarsky, 2016). In platform markets, algorithms continuously arrange, order, and rank product profiles and make personalized offers in exchanges based on the processing of large datasets (Callon, 2017; Glaser et al., 2021). The faster data accumulate, the faster profiles and products can be changed and (re)qualified (Kjellberg and Helgesson, 2006). They also translate, coordinate, and align the perspectives of different actors (Callon, 2017). In market studies terms, algorithms and the data infrastructures in which they are embedded have been deeply influencing actors' calculation processes. This has not gone unnoticed by regulators and market actors themselves, as our opening examples indicate: as algorithms become market actors in their own right, different market actors attempt to influence their "agency"—just as Amazon altered its algorithm to give higher visibility to product listings that yield greater profits for the company (Mattioli, 2019). On the other hand, to manipulate the algorithms in their favor, Amazon sellers might generate fake reviews, alter product lists with misleading information, and open several accounts to confuse the platform and their buyers to increase their sales (Stevens and Emont, 2018). Therefore, algorithms affect markets not only by transforming product

qualities and user interactions into a large amount of data and by influencing associated calculation processes, but they also deeply affect actor constellations and their power dynamics (Callon, 2017).

Technically speaking, algorithms are defined as "precise recipes that specify the exact sequence of steps required to solve a problem" (Glaser et al., 2021: 2), but the notions of "precise" and "exact" in this definition tend to obscure their technological and social embeddedness. Algorithms are technologically embedded as they interconnect with multiple other algorithms, which can obscure how they collect data from various sources and how their logics intertwine (MacKenzie, 2019). They are socially embedded in the contexts of their intended operations (Mackenzie and Vurdubakis, 2011; Ruckenstein, 2023), a context which in itself consists of shifting networks or assemblages of human and non-human actors (Glaser et al., 2021). It is the resulting continuous dynamic, we claim, that allows algorithms to become highly agentic in their own right, capable of channeling other actors' practices but crucially also deeply influencing and changing broader market dynamics—a capacity that we seek to capture in the notion of "agencing" (Cochoy et al., 2016; Geiger and Finch, 2016).

By drawing together market studies and critical algorithm studies, we identify four main roles of algorithms in shaping platform markets: market actor, target of market contestations, boundary object, and market systems connector. First, we show that platform algorithms can be considered as actors in their own right, gaining and affecting agency in that they continuously remake the market's agencements: they connect markets' objects, influence the calculation process, and shape market practices. Second, because of this increased agency, algorithms are central targets of market action: shaped and reshaped by actors, leading to constant reconfigurations in the "trials of strength" that constitute markets. Third, algorithms also need to be occasionally "pacified" (Callon, 2016) as boundary objects around which different actors gather to shape markets. Fourth, algorithms are connective market actors that can shape the broader market system as algorithms connect different platforms and distant actors together (Geiger and Kjellberg, 2021). We contribute to the market studies literature by uncovering how exactly algorithms may affect market dynamics. We further contribute to critical algorithms studies by offering more granularity to how algorithms affect some key processes influencing consumers' agency, such as consumer preferences, consumer welfare, and social equality. We close by outlining a market studies research programme for algorithmic studies in platform markets.

The agency of market devices

Conceptually, we situate our study in market studies, a research programme that over the past 20 years has accumulated knowledge on the theorizing of markets (Geiger et al., 2024). According to this perspective, markets are in a constant state of flux (Araujo and Spring, 2006; Kjellberg and Helgesson, 2006). Perceived as "socio-technical agencements," markets constantly enroll market objects in a complex web of comparisons and commensurations (Finch and Geiger, 2011; Çalışkan and Callon, 2010): actors translate their products into comparable goods based on the market criteria so that their goods are made qualifiable and exchangeable (Callon et al., 2002). Yet, market objects only become calculable as they connect with, define and stabilize other objects (Finch and Geiger, 2010). Calculation is based on singularizing and totalizing. Singularizing is a series of actions that translate an object of production into an object of a market, so that a good can be qualified (Callon, 2016; Callon et al., 2002; Callon and Muniesa, 2005). Meanwhile, markets also "totalize" by limiting the number of criteria to be taken into account through which goods can be related to each other (Callon, 1998). But as these characteristics are often forcefully limited, they have a tendency to re-enter the calculative process. Thus, calculating and recalculation, qualifying and requalifying is a

never-ending process in which a variety of market actors and devices are involved (Callon et al., 2002; Finch and Geiger, 2010; Kjellberg and Helgesson, 2006). Cochoy et al. (2016) have proposed the verb "agencing," from the French verb "agencer," to account for actions that seek to trigger and direct these dynamic processes, explaining that agencing both means producing specific agencements (that is, arranging market entities) and putting them in motion or "giving agency." Accordingly, "agencements denote sociotechnical arrangements when they are considered from the point [of] view of their capacity to act and to give meaning to action" (Çalişkan and Callon, 2010: 9). Market agencements are thus sociotechnical assemblages consisting of human actors but also non-human entities such as tools, objects, and algorithms. (Hardie and MacKenzie 2007).[1]

Market devices are centrally involved in these agencements. Market devices are "the material and discursive assemblages that intervene in the construction of markets" (Muniesa et al., 2007: 2). Market devices channel, influence, and sediment a market's calculative order. Research has highlighted both the agentic and sometimes unstable character of a large range of devices in shaping markets (Callon, 2017; Finch et al., 2017; Kaartemo and Nyström, 2021), extending to recent technologies. For example, the spread of cryptocurrency markets has presented a new challenge for traditional taxation and market regulation practices (Caliskan, 2022). Recent research has started to focus on the importance of data in market shaping, which may blur the distinction between calculative device and market object—a market assembled "through" data also often becomes a market "for" data (Cluley, 2020; Geiger and Gross, 2021). Research has further identified the role of digital market infrastructures such as http cookies to support knowledge production, capitalization, and coordination in markets (Mellet and Beauvisage, 2020).

As market devices, algorithms are embedded in many different markets, extending to markets for organ matching or online dating (Roscoe and Chillas, 2014), but they are in particularly widespread use in digital consumer-oriented platform markets (Airoldi and Rokka, 2022; Berman and Katona, 2020). In these markets, platform owners may design algorithms for recommending products to platform users (Zhong, 2023), promote relevant content to suitable users (Liu and Cong, 2023), or optimize advertising effectiveness (Malthouse et al., 2019). Platform owners might endeavor to use algorithms to order the market to exploit the algorithms in their own interests. While algorithms are reshaping broader market dynamics, they might also lead to more dominant positions of some actors and new forms of collusion (Li et al., 2021). Research has further shown that algorithms can result in more frequent price changes in online marketplaces, creating increasing pressure to order and regulate algorithmic markets (Cavallo, 2018; Chen et al., 2016).

The development and deployment of algorithms in markets are inseparable from the contexts surrounding them, involving multiple stakeholders who contribute in different ways, such as providing data, crafting algorithms, and adjusting parameters (Geiger, 2014; Gillespie, 2017). Platform users, for instance, supply the data that algorithms use, whereas data scientists and machine learning engineers handle the design and development of these algorithms and models. Additionally, legal professionals and regulators might be involved to ensure that ethical standards and legal requirements are met during the development process. As such, socio-technical agencements give algorithms their agency, but algorithms also re-order these socio-technical agencements. This agentic recursivity is amplified in platform markets since the data infrastructures in which algorithms are embedded form the very core "agencement" of these markets, which enable, consolidate, monetize, and regulate interactions between users and providers of content and services (Poell et al., 2021). As mentioned, algorithms in platform markets can create significant uncertainties in user interactions, market exchanges and value-creation practices due to their ever-changing logic (Pellandini-Simányi, 2023; Scaraboto and Fischer, 2024; Wood and Lehdonvirta, 2021). However, how exactly this happens can be difficult to grasp, not least given the interconnectedness between

different algorithms, which masks how they gather data from various sources and their logics become interconnected (MacKenzie, 2019). We claim that it is this dynamic, continuous re-configuration of the socio-technical agencements in which algorithms are involved that give them their agency—and that warrants heightened focus from market studies researchers.

Algorithm as a device and an actor in platform markets

To recall, in computer sciences, algorithms are seen as recipes or "computerized instruction sequence to solve certain tasks" (Lee and Helgesson, 2020). They are autonomous devices that automate decision-making processes without human interventions (Introna, 2016)—but, as mentioned, algorithms also take part in ill-defined and often shifting networks or assemblages of human and non-human actors that substantially exceed a "program" perspective of algorithms (Glaser et al., 2021). Critical algorithm studies have identified five main issues of concern around algorithms: opacity, power, social embeddedness, recursivity, and non-neutrality, which we will briefly discuss, leaning on Beer's (2013) and Airoldi and Rokka's (2022) classifications.

To start, the opacity of algorithms has attracted a lot of attention from social science scholars. While experts and regulators have been calling for greater algorithmic transparency, academics have endeavored to explain why such transparency is hard to achieve (Burrell, 2016). First, legal barriers for trade-secret protection prevent external stakeholders from gaining access to the source codes of algorithms (Burrell, 2016; Dourish, 2016). Second, comprehending and interpreting algorithms requires a high level of specialization, accessible mainly to a restricted professional group (Dourish, 2016). Third, even if auditing skills exist, the knowledge embedded in algorithms is challenging to assess since the algorithm mainly takes into account the unexpressed commonalities or proximities shared within vast amounts of training data: the algorithm might only reveal associations between data rather than "actually existing" relationships between the objects they represent (Dourish, 2016; Lee et al., 2019). Finally, opacity is inherent in the features of machine learning technologies. The original logic designed into algorithms by developers might change as machine learning trains the algorithm on training data (Burrell, 2016). Even organizations owning and/or using the algorithms might not be able to fully explain how they work (Hobbs, 2021). The opacity of algorithms can have significant impacts on our society as we tend to perceive what algorithms generate as the "normal" and "universal," despite the fact that the data these universals are based on are sometimes localized, particular, incomplete, or impure, and that connections among "real" objects might be only partially captured (Lee et al., 2019). Lee et al. use the example of financial algorithms to explain how they are substantially involved in "agencing" what are considered normal versus abnormal economic patterns, for instance for credit ratings, with significant consequences on market actors' scope of action (see also Pellandini-Simányi, 2023).

A second and related key feature of algorithms is their power. Algorithmic power, though often not apparent, is focused on determining what is shown and what stays hidden, and for whom (Bucher, 2012). In this regard, algorithmic power resembles (but sometimes also vastly exceeds) the power of traditional media in amplifying or suppressing certain windows into the world (Gillespie, 2017). The concept of an algorithm implies a sense of authority since algorithms are relied upon for their accuracy and impartiality (Beer, 2017). Their power does not only reside in directing computational processes; they also shape material realities as soon as social actors make decisions based on their results (Beer, 2013; Kitchin and Dodge, 2011). Algorithms thus possess an organizational power that shapes actors' range of possible actions and interpretations as well as their values and decisions (Beer, 2017; Mackenzie and Vurdubakis, 2011). Some social actors seek to become "complicit" with algorithmic power and leverage algorithmic frameworks for their own

purposes (Velkova and Kaun, 2021). For example, they try to understand the working mechanisms of algorithms and their influence on people's perceptions, and they try to draw on this understanding to change the mechanisms and influence so that the algorithms can perform based on the interests of specific actors (ibid.).

The third feature of concern is the algorithm's social embeddedness. This feature implies that we don't "see" algorithms as lines of code, but only encounter them indirectly as they integrate into daily routines (Beer, 2017). Algorithms are not "standalone little boxes, but massive ones with hundreds of hands reaching into them, tweaking and tuning, swapping out parts and experimenting with new arrangements" (Seaver, 2019: 419). Recent research has proposed that algorithms should be perceived as assemblages themselves, as algorithms possess inherently relational and distributed sociomaterial features that contribute to their making and remaking (D'Adderio et al., 2019; Glaser, 2017). This assemblage perspective signals the multitude of other actors—humans, rules, theories, data, other algorithms—that all contribute to algorithmic agencing. Moreover, this point highlights that even the most "autonomous" algorithmic agencement is not devoid of a larger context. Norms, values, or institutional logics are embedded into digital infrastructures, including algorithms and platforms (Hinings et al., 2018; Lee and Björklund Larsen, 2019; Wei et al., 2022), while algorithms themselves keep shaping what is institutionally taken for granted (Harmon et al., 2015). For instance, developers sometimes, and often unknowingly, impose a specific philosophical perspective on the functions of code and algorithms to address a specific relational problem (Kitchin and Dodge, 2011). A developer might develop a recommendation algorithm to recommend content that it predicts will be most liked by the majority, sometimes at the expense of niche interests or minority viewpoints (Liu and Cong, 2023). Such an algorithm reflects the philosophical perspective of utilitarianism, which the developer might not even be aware of.

The fourth feature of algorithms is recursivity, signaling that the results of a calculation process are incorporated into the input of a subsequent iteration in an algorithm (Beer, 2013). Certain algorithms are deliberately crafted to gather looped results, linking decision-making processes back onto themselves (ibid.). Research has shown that recursivity can have broad social implications (Airoldi and Rokka, 2022). For example, many widely used recommendation algorithms deduce user preferences by analyzing data on user traits and previous browsing history, enclosing users within self-validating feedback loops (Liu and Cong, 2023). As such, regardless of how random, peculiar, or imprecise the computation systems that refine the algorithm may be, they still lead to tangible results (Beer, 2017). In this way, algorithmic agency interacts with human agency, shaping our behaviors and thereby shaping themselves (ibid.). This is the case even when human actors deliberately disregard algorithmic decision-making. For example, when using recommendation systems for decision-making, customers frequently (and somewhat paradoxically) disregard the suggestions offered by the systems in favor of following their own instincts; Banker and Khetani (2019) have termed this phenomenon "algorithmic aversion." These decisions in turn might create systemic biases that can affect future customers, as data that feed back into the system (ibid.). Recursivity thus further complicates the predictability of algorithmic agencing.

The final and perhaps most discussed feature of algorithms is non-neutrality. For example, algorithms can favor mainstream best sellers (Fleder and Hosanagar 2009), promote higher-priced items (Banker and Khetani, 2019), and might even foster social inequalities in markets (Glaser et al., 2021; Sweeney, 2013). Due to a sense of certainty and expertise offered by algorithms, some users tend to overly depend on algorithms for decision-making, leading to risks to their own and others' well-being (Banker and Khetani, 2019). While algorithms are susceptible to biases stemming from bugs or coding mistakes (Diakopoulos, 2013), biases present in training and input data for algorithms can also lead to systematic reinforcement of discriminations in their output (Benjamin,

2019). These biases can also stem from developers—who have tended to be, according to Al-Amoudi (2023: 1243), 'unquestionably male, white, Western, geek and bourgeois'. Technical experts may not recognize their own subconscious biases and how these influence their development process, and they might resist external input from those who do not possess their technical knowledge (Danaher et al., 2017). According to Airoldi (2022), the reasoning that directs the actions of developers and computer programmers constitutes a form of culture embedded in various types of algorithmic agencements. For example, a field test of how an algorithm delivered ads promoting job opportunities shows that the advertisement was seen by fewer women than men purely as a result of the higher cost of showing ads to younger women (Lambrecht and Tucker, 2019). Algorithmic bias has triggered social resistance against algorithmic power through "repairing practices" such as deliberately "feeding" algorithms with data that are more diverse (Velkova and Kaun, 2021). Thus, non-neutrality both biases algorithmic agency and is a primary target of resistance against it.

Theorizing algorithmic agencing in markets

Drawing some of these insights from critical algorithm studies together with the notion of "agencing" in market studies, we theorize four roles that algorithms occupy in markets: as market actors, target of market contestations, boundary objects, and market systems connectors.

Market actors

First, algorithms on platforms are market actors in that they do not only connect markets' objects such as products, data, buyers, and producers, but they do so with tangible effects on other market actors. Algorithms influence calculations and shape market practices such as collecting data from sellers, ranking their products and making personalized offers (Yoon et al., 2021). These calculative practices are often based on a belief in the objectivity, neutrality, and rationality of its working process (Beer, 2017)—quite simply, most platform users "trust" algorithmic outputs. Algorithms also exert indirect power on other market actors by framing their possibilities of actions and understanding (Beer, 2013). As outlined, the European Commission's investigation into Amazon's e-commerce platform revealed that the company used its algorithms to preferentially select products from its own retail business and marketplace sellers that use Amazon's logistics and delivery services into its Buy Box (European Commission, 2022). Thus, algorithms as market actors shape platform users' practices by making some information visible and other information invisible (Bucher, 2012). In some cases, users might not even be aware of the existence of algorithms influencing their decision-making (Gran et al., 2021).

Of course, from an actor-network perspective, agency and autonomy are relative terms: as algorithms constantly enroll and exclude markets' objects, they work relationally—that is, they are also targets of others' actions. For example, on Amazon, multiple algorithms work together to identify, compare, and rank products on product pages, include competitors' data to change the prices of these products, and exclude or delete fake reviews and sellers with misconduct (Stevens and Emont, 2018). But algorithms can also work against each other in complex "interaction orders," which may even include deception and "spoofing" (MacKenzie, 2019). For example, in a trading platform, traders use algorithms to create fake orders. These orders are intended to give false signals to other trading algorithms, manipulating market prices or the behavior of other traders' algorithms. Agencing is, therefore, an ongoing process of arranging and adjusting the elements of a practice, while also adapting to the agency of other elements involved in the same practice (Ryan et al., 2023).

Interactions between algorithms further strengthen their autonomy and increase their opacity, since nobody can accurately explain how they interact, including the regulators and the platform owners themselves. For example, during the EU investigations into Amazon's Buybox, Amazon confessed that it cannot describe for certain how its algorithms work since they are not simply a set of "if" or "then" statements (Hobbs, 2021). If nobody can understand them, nobody can control how they "make" autonomous decisions—and how they might agence markets as a result. Importantly, beyond influencing actors' calculative processes, the algorithms also represent platform markets to other actors as they arrange market objects on behalf of platform owners. However, such a representation might be distant from reality or even be misleading, since an algorithm that represents a market assembled "through" data may blur the distinction between calculative device and market object (Cluley, 2020). Geiger and Gross (2021), for instance, investigate the platform market for consumer genomics tests, in which one of the biggest draws for potential customers is the calculation of ancestry relationships—including what percentage of "Neanderthal" DNA a person possesses. Yet, how these percentages are calculated remains not only a trade secret but more importantly changes in relation to who submits to the DNA analysis in the first place.

Targets of market contestations

As algorithms are embedded in larger socio-technical market agencements, they can also become targets of market contestations. The designers of the algorithms constantly seek to shape the algorithms by offering training data and design reasoning, thereby "theorizing" the algorithm's agency (Glaser et al., 2021)—but so do those who seek to contest or undermine these algorithms (Cotter, 2019; Kear, 2017). The more, however, algorithms become a battleground of different agencies, as in our example of traders feeding "false" data to competitor algorithms, the less stakeholders will trust their results or predictions. Importantly, due to opacity, users in platform markets might try to influence markets by interacting with algorithms, but how the algorithms work is not transparent to them, which makes such battlegrounds epistemological guessing games. And the less trust in algorithms market actors have, the less they will follow algorithmic recommendations. For example, users might try to "outguess" the algorithm and make decisions contrary to those recommended, or they will disregard the market representation made by the algorithms in the decision-making process. As these decisions in turn flow back into markets (due to the feature of *recursivity*), they will further bias market actors' calculations and reduce trust.

For example, Shopify was sued by some textbook publishers who claimed that Shopify is benefiting from products that are pirated or infringe on their trademarks (Deschamps, 2022). They claimed that Shopify enables these sellers to use its software to sell products that are nearly identical to their products. However, the company denied these claims and asserted that they frequently take down relevant product listings. As the case reveals, due to the opacity of algorithms, Shopify and these textbook publishers have each provided their own interpretations of the algorithmic results, leading to a bitter dispute over market practices. The case has been settled between Shopify and these publishers (ibid.).

Although the exact agencing of the algorithms often remains hidden from their users (including sellers, buyers and consultants), they may try to understand it based on their repeated interactions with the algorithms and their communications with peers. They develop folk theories about the logics of algorithms, adapting their behaviors accordingly (Eslami et al., 2016). For example, if hosts on Airbnb platform suspect that their rankings will drop if they use the platform's dispute resolution system, they become hesitant to use the system (Cheng and Foley, 2019). Other hosts find out that they have better business performance if they decline requests less frequently but change

their description lists more frequently, but these are based on their intuition after they experimented with the algorithms (ibid.). Kear (2017: 346) studied the folk theories developed by a peer lending group that tries to "game" algorithmic credit ratings by feeding it with credit-friendly data, thus leveraging the algorithm's "blind spots and biases through compensatory and transgressive data-generation practices." Further, in contrast with the assumption that data-driven algorithms can weaken small companies' advantages (OECD, 2017), knowledgeable users will exploit their knowledge of the algorithms by offering false data to influence the calculation process. eBay sellers may post fake reviews to enhance their visibility in the marketplace (Smithers, 2020). Hosts on AirBnB created fake listings by misrepresenting the listings' addresses (Humphries, 2023).

In these contestations, the platform transforms into explanatory spaces, or exploratoriums (Callon, 2021) where actors seek to investigate algorithmic agency and shape it to their needs. However, such algorithmic contestations can create power imbalances, for instance, to the detriment of new platform users, since the understanding of and ability to manipulate algorithms depends on their skill levels and experience (Gran et al., 2021). In addition, with algorithms' logics being emergent and recursive, users' every interaction with algorithms to understand their functioning also becomes input data. In other words, the algorithm treats these explorations as authentic expressions of "use" (Cotter, 2019).

Importantly, algorithms can also become battlegrounds for social justice. By highlighting potential algorithmic misfires, especially those related to the features of non-neutrality or power—for instance, in racial or gender biases—those contesting an algorithmic status quo may be able to demand that these overflows be brought back into the market and "corrected." This could be done by highlighting the need to diversify training data used or to correct social assumptions embedded in codes (Pellandini-Simányi, 2023). Regulators have attempted to implement auditing processes to reduce risks of algorithmic "misbehavior," but the opacity of the algorithms weakens the impacts of such governance. For example, the European Commission has recently mandated 19 platforms, including Amazon and Booking.com, to disclose "algorithmic risks" as part of the Digital Service Act. Their plans for reporting and managing these risks will undergo independent auditing and supervision by the European Commission (Lomas, 2023). However, since a report has mentioned that investigators faced difficulties in opening the "black box" of codes (Hobbs, 2021), the effects of these future auditing processes of their algorithms are uncertain. To pacify different stakeholders, platform owners have offered to detect these misconducts, improve the reasoning of the algorithms, and expel market actors with misconducts. For example, AirBnB has expelled hosts with fake listings and planned to improve its algorithms with AI technologies to verify these listings (Koenig, 2023). Such market contestations—or, with Callon, trials of strength—continuously reconfigure both the algorithms themselves and the power distribution in markets. What is supposed to be a relatively stable calculative assemblage becomes not only highly unstable but also continuously contested.

Boundary objects

This leads us to the third role that algorithms occupy in markets: namely, as boundary objects. Boundary objects are negotiation devices: they sit at the boundary between different actors' realms of action and meanings and both facilitate and are utilized in interactions with others (Star and Griesemer, 1989). In markets, boundary objects are designed to perform boundary tasks by translating, coordinating, and aligning the perspectives of different actors who use them in their interactions and negotiations over what is inside and what is outside a market (Finch and Geiger, 2010). On the one hand, understanding algorithms as boundary objects allows to reveal the constant

battles between platform owners and sellers, as both of them try to take actions to leverage algorithms to influence each other—that is, shift market boundaries in their favor (Finch and Geiger, 2010). But unlike our reflections above around algorithms as targets of market contestations, for algorithms to assume the role of a boundary object, they also need to be at least occasionally pacified—market actors cannot negotiate over boundaries that are too unstable. Thus, despite emergent logics and recursive processes, algorithms need to stabilize enough to "peacefully" coordinate market actors' intentions and requirements in order to stake out markets' boundaries. For example, Google's algorithms fulfill the needs of merchants to make customized advertising contents and the needs of the platform owner Google to rank and order the contents in its search engine's results (see Rangaswamy et al., 2009); it thus translates between platform market actors (relatively) predictably. However, such a "pacified" state might only remain peaceful for a short period due to boundary battles. While Google's algorithms were considered to be fair by many users for a long time, the EU recently accused Google of using its algorithms to enhance the appearance of its own services with appealing characteristics in the search results and employing algorithms that tend to suppress the outcomes of competitors (Chee, 2023a).

Algorithms also need to be robust enough to maintain a common identity for both sides as the algorithms represent the reputation of the marketplace. Since such a common identity is inscribed in the algorithms, platform owners might need to manage their algorithms accordingly. For example, it is reported that merchants on Alibaba's Aliexpress marketplace are feeding its algorithms with illegal content or counterfeit products (Chee, 2023b). These illegal contents might have undermined the common identity of the merchants and Aliexpress as a marketplace for legal trade. Since algorithms as boundary objects delineate competition dynamics as market actors interact and compete with each other within the bounds of algorithms, it is important to manage the algorithms in order to maintain a common identity of the platform marketplace for fair competition. However, given the likelihood of platform-internal contestations, as highlighted above, it is challenging to keep a common identity and boundary due to algorithms' opacity and recursive processes.

Market systems connectors

Emphasizing algorithms' relational quality, we finally propose that algorithms can be conceptualized as market systems connectors that can affect the market system beyond immediate buyer-seller exchanges. Algorithms connect individual markets into market systems (MacKenzie, 2019): they constantly collect data from different platforms and take actions to change the elements in exchanges, such as prices, product images, and distribution modes, thereby reshuffling not simply one market's calculative apparatus but in fact connecting and re-arranging entire market systems. For example, based on the data collected from other competitor platforms, the algorithms on Amazon enable third-party sellers to adjust their pricing to match the lowest prices available from competitors, which will have knock-on effects on the original market—and is also a form of market collusion (Competition and Markets Authority, 2018). Additionally, algorithms also constantly shape the actor-network by reconfiguring market pathways. For example, based on the data collected from different sellers and buyers, the algorithms on the platform can identify the most suitable match between them (e.g., Zhong, 2023). Algorithms finally change institutional intermediaries by forming new parameters to represent the market and its transactions, for instance, to regulators (Geiger and Kjellberg, 2021). For example, the pricing data collected by the algorithms across different platforms indicate the sellers' product competitiveness in the market (Zhong, 2023). Based on the data, platform owners, their competitors and sellers on different platforms take actions in market exchanges. They can even collude to gain advantages in market exchanges across multiple

markets by developing algorithms to monitor actions across markets, thus utilizing the algorithm's systemic effects to escape stringent regulation (see Li et al., 2021). Therefore, algorithms connect different platform markets and trigger dynamic changes in how these markets interconnect, leading to broader systemic and even institutional changes.

However, yet again how exactly such processes are enacted likely remains a black box for market actors since these systemic changes result from the collective interactions among multiple algorithms and data infrastructures (MacKenzie, 2019), and the logic and data used to develop these algorithms and how these are shaped through systemic interaction are never completely transparent. This connectivity in turn compounds the opacity of algorithms since they receive data from different sources and their logics are intertwined (ibid.). In addition, since every algorithm is influenced by different social factors in their own contexts (Beer, 2017), social embeddedness might lead to unexpected changes in an algorithm due to a distant social factor. Such changes can further create ripple effects due to the connectivity of algorithms.

A research programme for algorithmic market studies

We utilize our fourfold categorization above as a conceptual basis to formulate a research programme for market studies concerning algorithmic agencing. First, with this paper, we emphasized the importance of investigating the agency of algorithms in shaping platform markets. As *market actors*, algorithms can shape and transform markets. While this seems to be the case for any market actor and market device, algorithms do so with one crucial difference: the speed at which they enact this "agencing." Based on an instant analysis of large amounts of data, algorithms as market actors vastly accelerate market dynamics, literally to the "speed of light" (MacKenzie, 2021). Going forward, market studies research should examine in more detail the temporalities of algorithmic agencing and its effects, given the four roles we identified above. The speed of algorithms' influence on market dynamics creates significant instabilities that affect not only the market actors' conduct but also the algorithms themselves, which change shape all while being the target of others' (in the case of other algorithmic actors, often equally rapid) agency; with the result that algorithmic market actors, so to speak, are constantly ahead of their human or non-algorithmic counterparts. Notwithstanding the research challenges inherent in studying processes that by their nature are slippery, (kept) obscure, and are deeply buried in markets' digital infrastructures (MacKenzie, 2019; Mellet and Beauvisage, 2020), we strongly encourage future research to examine how algorithms change the temporality of market dynamics and how this change, in turn, affects the interactions and power distributions between different actors in specific platform markets. Ideally, this work should be sensitive to the different temporalities in play in this process, where the instantaneity of algorithmic agencing may clash with and disturb slower or more settled market practices, such as, for instance, regulations. Relatedly, future research also needs to investigate how algorithms unsettle regulations or generate new market norms over time and how regulators develop new measures to regulate algorithm-driven markets. As we have indicated through our brief examples in this paper, court materials may prove vital in retracing the controversies and contestations that algorithms may give rise to and any resulting effects on market norms and infrastructures; they might also provide rare glimpses into normally opaque processes of algorithmic agencing.

Future algorithmic market studies research may follow the lead we have given in this paper and shift toward the analysis of broader algorithmic agencements and the actions market actors take to influence those, especially in platform markets. Researchers could, for instance, explore the difficulties involved in representing markets that are hyper-dynamic and full of "opaque" market actors (Kjellberg and Helgesson, 2006). Algorithms provide a dense constellation of devices that

constantly identify market objects, associate these markets' objects with one and another, and select, rank and order these connecting market objects. However, the connections between market objects organized by algorithms sometimes do not reflect the representations that market actors or researchers may have of a market; to recall, to algorithms, market objects are represented by data and their relationships are represented through the connections between sets of data, not those of "real" objects (e.g., Dourish, 2016; Lee et al., 2019). Thus, an "algorithmic" market representation may be based on relationships that do not reflect what "non-algorithmic" market actors perceive the market to be. Users' explorative practices and other data not directly emanating from market exchanges may further distance the representation of markets from any market "reality" otherwise perceived; future research could thus also investigate how different versions of markets are affected by these recursive algorithmic processes. Therefore, research should study how multiple versions of markets generated by algorithms may be represented and by whom and how this, in turn, affects market dynamics, the development of market norms, and scopes of action.

Second, bringing to bear their thinking of market agencements as relational, socio-technical, and embedded in different normativities (Çalışkan and Callon, 2010), market studies researchers may also be proactive in helping to formulate future policies for platform markets governed by algorithms. Since algorithms are *the targets of market contestations*, other connected actors and devices also influence the development process of algorithms, thereby shaping market dynamics. When elucidating the systemic effects of algorithms pertinent to policy formulation, market studies researchers are encouraged to meticulously delineate the interconnected entities and their dynamic interactions within socio-technical assemblages. This would be particularly helpful in cases where the opacity of algorithms constrains regulators' capabilities in understanding, investigating, and auditing the market norms generated by algorithms in order to avoid what might be called "algorithmic capture," paraphrasing Finch et al. (2017)'s notion of "material capture." By viewing algorithms as the targets of market contestations, further research should focus on how the intransparent nature of algorithms affects setting market standards and representing markets, thereby potentially affecting social justice concerns in markets—for instance, questions of who is "invited" into a market agencement and who is excluded, how differential pricing affects accessibility etc. With rapid advances in Artificial Intelligence, such algorithmic social justice concerns become more vital than ever (Gross, 2023). Such research may also fruitfully draw on and contribute to constructivist literature on transparency in organization studies (e.g. Geiger and Bourgeron, 2023; Glaser, 2017; Glaser et al., 2021) to critically interrogate in whose interest different ways to "devise" opacity and transparency in platform markets are.

Third, the process of boundary work (Cochoy et al., 2016; Gieryn, 1983; Stigzelius et al., 2018) involves multiple stakeholders from different fields, such as in our case platform owners or users across adjacent platforms. As *boundary objects* in platform markets, algorithms may become both the touchpoint and the battlefield between rival market agencements. Yet at the same time, "truces" in these battles are necessary, as we argued—and they are only possible if algorithms are periodically pacified and allow actors to stake claims over market "territories" and boundaries. Therefore, when studying the impact of algorithms on market shaping, future research should include their role into demarcating the market network. The agentic quality of the algorithms substantially increases their complexity as boundary objects as they are constantly changing themselves, thereby potentially destabilizing the market's "territory." Future research needs to urgently pay attention to the dynamics between "activation" and "pacification" (Callon, 2016) when studying their roles in shaping platform market boundaries. Additionally, as mentioned above, the recursivity of algorithms increasingly diversifies the perceptions of a market, thereby adding obstacles in maintaining a market agencement's identity to external actors. Future research may thus

seek to empirically explore how recursivity affects their roles as boundary objects and how platform owners utilize algorithms to demarcate spaces for their market networks.

Fourth, as *market systems connectors*, algorithms do not only connect different actors in the marketplace or demarcate territory across different platforms, but they may also influence broader market systems. The algorithms constantly collect and analyze data from different platforms and take actions to change the exchange elements, actor network, and institutional transmitters in the marketplace (MacKenzie, 2019). Such processes powered by algorithms connect different platform markets, and different platform markets are interacting with each other through these algorithms. It is clear that these coordination processes might go far beyond individual platforms and algorithms. In the case of digital consumer platforms, for instance, though they are in competition in some territories, the combined market power of Alibaba and Amazon has repercussions across a broad swathe of market systems. The connection points between disparate entities can be multiplied beyond the algorithms; they may also include other actors from different fields, such as legislators, associations, and companies from a variety of industries, which become "enrolled" through algorithmic agency and contribute to adjusting or extending the market's frames of reference. Therefore, when studying algorithmic markets, researchers need to capture a broad range of actors, some of whom may at first appear distant. Particularly, they should identify the actors who are directly involved in the agencing process and the connection points between these actors and those that are less proximately involved, as the connection points might imply the actors' indirect influence on the algorithms. Additionally, the functioning of one algorithm on one platform may depend on another algorithm on another platform, so the interdependent processes between algorithms may hinder other actors from uncovering algorithmic reasoning, thereby further compounding algorithms' opacity. Consequently, algorithms that are heavily networked tend to have higher autonomy to make decisions and enhanced agency since it becomes more and more challenging to understand and control them. We encourage future research to investigate how interlinked processes between algorithms in the broader market system affect the relative agency of algorithms and other actors. Further research may also explore how the interpretability of algorithms may be increased to counterbalance algorithmic power. As other smart technologies such as the Internet of Things and blockchain also connect actors in broader market systems (Pardo et al., 2022; Wei and Pardo, 2022), future research should study how these connected smart technologies take part in and are influenced by algorithmic agencing—that is, they should study algorithmic battlefields at a systemic level. This research also needs to prominently include ethical considerations around algorithmic power and agency. Clearly, the "politics" of these devices matter (Al-Amoudi, 2023), in markets and beyond.

Finally, bringing our four dimensions of algorithmic agency in markets together, we propose a paradigm of dualism: as market actors, algorithms can both facilitate and suppress market agency. In market contestations, they are both the battlefields and the weapons that other market actors (including other algorithms) will wield. As boundary objects, algorithms can translate, coordinate, and align the perspectives of different actors effectively and efficiently due to their instant analysis of big data, but they also serve to distance some actors from certain market agencements. Furthermore, as market systems connectors, algorithms might further amplify these contestations to different platform markets or even beyond, into non-market forums. However, we still have limited knowledge regarding how these respective dualistic effects manifest and interact. We strongly encourage future research to explore the triggers and boundary conditions of their manifestations and interactions. A contingency approach may also usefully reveal how algorithms shape market dynamics in specific markets. Such an understanding can be linked to prominent topics in the consumer domain, such as how consumer preferences are shaped, how social inequalities are

formed, and how consumer welfare is affected, thereby enriching the literature of the critical studies of algorithms.

Conclusion

Algorithms are central actors in platform markets, rapidly and constantly shaping these markets and in turn being continuously shaped and reshaped by multiple other market actors, including other algorithms. We argue that the dynamics they introduce in market processes represent a substantial and important shift in our understanding of market agency. In particular, their intransparent and networked nature and the speed of change they introduce to markets bring challenges to market actors to control them and to market studies researchers to investigate the agentic distribution in these market agencements. Market actors including platform owners, platform users, regulators, and researchers should be aware of the multiple agentic roles of algorithms in platform markets when attempting to represent and analyse market dynamics. Our paper provides urgently needed conceptual guidance to market actors on how to interact with each other and with algorithms in platform markets. We contribute to the market studies literature by uncovering why and how algorithms affect market dynamics. We further propose a dualistic lens on algorithms' market agency. Our conceptualization of algorithms' roles also reveals how the key features of algorithms in critical algorithms literature manifest in shaping platform market dynamics. Such an understanding of the roles of algorithms offers more granularity to how other market actors' agency is affected by algorithms in platform markets and beyond. Our proposed research programme now needs to be filled with empirical investigations to explore the many ways in which algorithms "agence" markets to guide policy, managerial decision-making, and academic research.

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Note

1. In this paper, we use the terms markets and market agencements interchangeably.

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